

Alamy Stock Photo; **bottom left** Amar and Isabelle Guillen - Guillen Photo LLC/Alamy Stock Photo; **top right** Eldon H. Newcomb; **bottom right** Science Photo Library/Alamy Stock Photo; **33.10** CDC; **33.11** Eye of Science/Science Source; **33.11** Eye of Science/Science Source; **33.12** M. I. Walker/Science Source; **33.13** Holger Herlyn, University of Mainz, Germany; **33.14a** blickwinkel/Alamy Stock Photo; **33.14b** Image Quest Marine; **33.16** Image Quest Marine; **33.17a** Lubos Chlubny/Fotolia; **33.17b** Terry Moore/Stocktrek Images/Alamy Stock Photo; **Scientific Skills Exercise** Christophe Courteau/Water Rights/Alamy Stock Photo; **33.18** Andrew J. Martinez/Science Source; **33.20 top** Mark Conlin/VWPics/Alamy Stock Photo; **center** Photonimo/Shutterstock; **bottom** SeaTops/Alamy Stock Photo; **33.21 left** Dave Clarke/Zoological Society of London; **right** The U.S. Bureau of Fisheries; **33.22** Fredrik Pleijel; **33.23** Wolcott Henry/National Geographic; **33.24** Astrid Michler; Hanns-Frieder Michler/Science Source; **33.25** Wayne Taylor/The AGE/Fairfax Media via Getty Images; **33.26** London Scientific Films/Oxford Scientific/Getty Images; **33.27** Power and Syred/Science Source; **33.28** Dan Cooper; **33.29b** Courtesy of Sean B. Carroll; **33.31** Mark Newman/Frank Lane Picture Agency; **33.32 top** Tim Flach/The Image Bank/Getty Images; **center** Andrew Syred/Science Source; **bottom** Reinhard Hölzl/ImageBROKER/AGE Fotostock; **33.34a** Premaphotos/Nature Picture Library; **33.34b** Tom McHugh/Science Source; **33.36** Maximilian Weinzierl/Alamy Stock Photo; **33.37** Peter Herring/Image Quest Marine; **33.38** Peter Parks/Image Quest Marine; **33.40** André Skonieczny/F1online digitale Bildagentur GmbH/Alamy Stock Photo; **33.41a, b, d, e** Cathy Keifer/Shutterstock; **33.41c** Jim Zipp/Science Source; **33.42** Archaeognatha Kevin Murphy; **Zygentoma** Denis Crawford/Alamy Stock Photo; **Coleoptera** Premaphotos/Nature Picture Library; **Diptera** Bruce Marlin; **Hymenoptera** John Cancalosi/Nature Picture Library; **Lepidoptera** Hans Christoph Kappel/Nature Picture Library; **Hemiptera** Dante Fenolio/Science Source; **Orthoptera** Chris Mattison/Alamy Stock Photo; **33.43** Andrey Nekrasov/Image Quest Marine; **33.44** Daniel Janies; **33.45** Jeff Rotman/Science Source; **33.46** Louise Murray/robertharding/Alamy Stock Photo; **33.47** Jürgen Freund/Nature Picture Library; **33.48** Hal Beral/Corbis; **p. 717** beetles Lucy Arnold.

Chapter 34 **34.1 top to bottom** Derek Siveter, Tom McHugh/Science Source, Gino Santa Maria/Shutterstock; Digital Vision/Photodisc/Getty Images, Arnaz Mehta, Tom McHugh/Science Source, Rolf Nussbaumer Photography/Alamy Stock Photo, Visceralimage/Fotolia; **34.4** Natural Visions/Alamy Stock Photo; **34.5c** Ethan Daniels/Stocktrek Images/Alamy Stock Photo; **34.8** Tom McHugh/Science Source; **34.9** Marevision/AGE Fotostock; **inset** Hartl/blickwinkel/Alamy Stock Photo; **34.10** Junyuan Chen/Nanjing Institute of Geology and Palaeontology, Chinese Academy of Sciences; **34.14** Field Museum Library/Premium Archive/Getty Images; **34.15a** Gino Santa Maria/Shutterstock; **34.15b** Masa Ushioda/Image Quest Marine; **34.15c** RGB Ventures/SuperStock/Alamy Stock Photo; **34.17** tuna James D. Watt/Image Quest Marine; **lionfish** Teigler/blickwinkel/Alamy Stock Photo; **sea horse** George Grall/National Geographic; **eel** Fred McConnaughey/Science Source; **34.18** Reprinted by permission from Macmillan Publishers Ltd. From: The oldest articulated osteichthyan reveals mosaic gnathostome characters. M. Zhu. *Nature*. 2009 Mar 26;458(7237):469-74. doi: 10.1038/nature07855. Fig. 2; **34.19** Arnaz Mehta/SeaPics; **34.20** fossil, ribs, scales Ted Daeschler/Academy of Natural Sciences/Vireo; **fin** Kalliopi Monoyios Studio; **34.22a** Alberto Fernández/AGE Fotostock/Alamy Stock Photo; **34.22b** Anneka/Shutterstock; **34.22c** Zeeshan Mirza/ephoto corp/Alamy Stock Photo; **34.23a** DP Wildlife Vertebrates/Alamy Stock Photo; **34.23b** FLPA/Alamy Stock Photo; **34.23c** John Cancalosi/Photolibary/Getty Images; **34.24** Hinrich Kaiser, Victor Valley College; **Problem-Solving Exercise** Joel Sartore/National Geographic; **34.27** Nobumichi Tamura; **34.28** Chris Mattison/Alamy Stock Photo; **p. 736** Sereno Paul Sereno Fossil Lab; **34.29a** Natural Visions/Alamy Stock Photo; **34.29b** Lee T. Matt; **34.29c** Nick Garbutt/Nature Picture Library; **34.29d** Juniors Bildarchiv/AGE Fotostock; **34.29e** Carl & Ann Purcell/Corbis NX/Getty Images; **34.30a** Visceralimage/Fotolia; **34.30b** The Natural History Museum/Alamy Stock Photo; **34.32** Boris Karpinski/Alamy Stock Photo; **34.33** DLILLC/Corbis/VCG/Getty Images; **34.34** Mariusz Blach/Fotolia; **34.35** The Africa Image Library/Alamy Stock Photo; **inset** mychicport/Shutterstock; **34.36** Gianpiero Ferrari/Frank Lane Picture Agency Limited; **34.39** Clearviewstock/Shutterstock; **inset** Commonwealth Scientific and Industrial Research Organization; **34.40a** John Cancalosi/Alamy Stock Photo; **34.40b** Martin Harvey/Alamy Stock Photo; **34.40c** Rick & Nora Bowers/Alamy Stock Photo; **34.43** ImageBroker/Alamy Stock Photo; **34.45a** Kevin Schafer/AGE Fotostock; **34.45b** J & C Sohns/Picture Press/Getty Images; **34.46a** Morales/AGE Fotostock; **34.46b** Tim Laman/NaturePL; **34.46c** T.J. Rich/Nature Picture Library; **34.46d** E.A. Janes/AGE Fotostock; **34.46e** Martin Harvey/Photolibary/Getty Images; **34.48** T. White/David L. Brill Photography; **34.49a** John Reader/Science Source; **34.49b** Mauricio Anton/Science Source; **Scientific Skills Exercise** Golfx/Shutterstock; **34.50** Danita Delimont/Alamy Stock Photo; **34.52a** Erik Trinkaus; **34.52b** Tom Higham, University of Oxford; **34.53** David L. Brill Photography; **34.54** From: *Homo naledi*, a new species of the genus *Homo* from the Dinaledi Chamber, South Africa. L. R. Berger et al. *eLife* 2015;4:e09560. Fig. 6; **34.55** C. Henshilwood; **p. 756** animal Tony Heald/Nature Picture Library.

Unit Six Interview Courtesy of Dennis Gonsalves.

Chapter 35 **35.1 tree** Raimund Linke/Photodisc/Getty Images; **seedling** Beata Becia/Shutterstock; **leaf cross section** P&R Fotos/AGE Fotostock/Alamy Stock Photo; **chloroplasts** John Durham/Science Source; **tube-shaped cells** Science Photo Library/Alamy Stock Photo; **root hairs** Scenics & Science/Alamy Stock Photo; **35.3** Jeremy Burgess/Science Source; **35.4** buttress roots Karl Weidmann/Science Source; **prop roots** Natalie Bronstein; **beet** Rob Walls/Alamy Stock Photo; **pneumatophores** Björn Svensson/AGE Fotostock/Alamy Stock Photo; **strangling roots** Dana Tezarr/Photodisc/Getty Images; **35.5 top** Maureen Spuhler/Seelevel.com; **center** Dorling Kindersley Ltd/Alamy Stock Photo; **bottom** Toshihiko Watanabe/Alfo/Alamy Stock Photo; **35.7 tendrils** Neil Cooper/Alamy Stock Photo; **spines** Martin Ruegner/Photodisc/Getty Images; **storage leaves** Dmitry Skorobogatov/123RF; **reproductive leaves** Godonova Tatiana/Shutterstock; **Scientific Skills Exercise** Dorling Kindersley Ltd/Alamy Stock Photo; **35.9** Steve Gschmeissner/SPL/AGE Fotostock; **35.10** parenchyma M I (Spike) Walker/Alamy Stock Photo; **35.10** collenchyma Keith Wheeler/Science Source; **scleireid** Graham Kent/Pearson Education; **fiber** Graham Kent/Pearson Education; **tracheids and vessels** N.C. Brown Center for Ultrastructure Studies; **sieve-tube element (TEM)** From: *Plant Cell*

Biology on DVD: Information for students and a resource for teachers Springer-Verlag 2009, by B Gunning; **sieve-tube element (LM)** Ray F. Evert; **sieve plate** Graham Kent/Pearson Education; **35.13** From: ABA-mediated ROS in mitochondria regulate root meristem activity by controlling PLETHORA expression in *Arabidopsis*. Yang L. *PLoS Genet.* 2014 Dec 18;10(12):e1004791. doi: 10.1371/journal.pgen.1004791. eCollection 2014 Dec. Figure 6G; **35.14a, b** Ed Reschke; **35.14a center** Chuck Brown/Science Source; **35.15, 35.16** Michael Clayton; **35.17, 35.18** Ed Reschke; **35.20 left** Michael Clayton; **right** Alison W. Roberts; **35.23** University of Southern California; **35.25** Edu Boer, NVWA, NL; **35.26** From: Natural variation in *Arabidopsis*: from molecular genetics to ecological genomics. D. Weigel. *Plant Physiol.* 2012 Jan;158(1):2-22. doi: 10.1104/pp.111.189845. Epub 2011 Dec 6. Fig. 1; **p. 776** Chory Micheline Pelletier/Courtesy of Joanne Chory; **35.28** From: Microtubule plus-ends reveal essential links between intracellular polarization and localized modulation of endocytosis during division-plane establishment in plant cells. P. Dhonukshe. *BMC Biol.* 2005 Apr 14;3:11. Fig4B; **35.28 (inset)** B. Wells and K. Roberts; **35.29** From: The making of a compound leaf: genetic manipulation of leaf architecture in tomato. D. Hareven. *Cell.* 1996 Mar 8;84(5):735-44. Fig. 1; **35.30** From: A common position-dependent mechanism controls cell-type patterning and GLABRA2 regulation in the root and hypocotyl epidermis of *Arabidopsis*. C. Y. Hung et al. *Plant Physiol.* 1998 May;117(1):73-84. Fig. 2g; **p. 778** Benfey Jie Huang/Duke University; **35.31** Lawrence Jensen; **35.32** From: Genetic interactions among floral homeotic genes of *Arabidopsis*. J.L. Bowman, DR Smyth, EM Meyerowitz. *Development.* 1991 May;112(1):1-20; Fig. 1A; **p. 780** Walbot Courtesy of Virginia Walbot; **p. 782** woody eudicot From: Anatomy of the vessel network within and between tree rings of *Fraxinus lanuginosa* (Oleaceae). P. B. Kitin et al. *American Journal of Botany.* 2004;91:779-788. Fig. 1; **p. 783** tea leaves Volodymyr Burdiak/Shutterstock; **tea leaf cross section** Keith Wheeler/Science Source; **iris leaves** Rob Stark/Shutterstock; **iris leaf cross section** www.willemsmicroscope.com; **bicycle** Janet Horton/Alamy Stock Photo; **Hakea purpurea** Biophoto Associates/Science Source.

Chapter 36 **36.1** velislava/Alamy Stock Photo; **36.3** Rolf Rutishauser and Evelin Pfeifer; **p. 790** plant Nigel Cattlin/Alamy Stock Photo; **36.8** Benjamin Blonder and David Elliott; **36.10** Scott Camazine/Science Source; **36.13** AGE Fotostock/Alamy Stock Photo; **36.14** Power and Syred/Science Source; **36.15** leafless ocotillo mike lane/Alamy Stock Photo; **green ocotillo** Rick & Nora Bowers/Alamy Stock Photo; **ocotillo close-up** Mint Images/SuperStock; **oleander cross section** Natalie Bronstein; **blossom** Sutidas/Shutterstock; **cactus** Danita Delimont/Alamy Stock Photo; **36.18** M. H. Zimmerman/Harvard Forest; **36.19** From: A coiled-coil interaction mediates cauliflower mosaic virus cell-to-cell movement. L. Stavolone et al. *Proc Natl Acad Sci U S A.* 2005 Apr 26;102(17):6219-24. Epub 2005 Apr 18. Fig. 5c; **p. 802** Zambryski Noah Berger Photography; **p. 804** forest Catalin Petolea/Alamy Stock Photo.

Chapter 37 **37.1** Visuals Stock/Alamy Stock Photo; **37.2** ARS/USDA; **37.4** National Oceanic and Atmospheric Administration (NOAA); **37.5** Menlo Park/USGS; **37.6** Kevin Horan/The Image Bank/Getty Images; **p. 811** Coruzzi Courtesy of Gloria M. Coruzzi; **37.8** healthy View Stock RF/AGE Fotostock; **nitrogen-deficient** Guillermo Roberto Pugliese/International Plant Nutrition Institute (IPNI); **phosphorus-deficient** C. Witt/IPNI; **potassium-deficient** M.K. Sharma and P. Kumar/IPNI; **Scientific Skills Exercise** Nigel Cattlin/Science Source; **37.9** lichen David T. Webb, University of Montana; **lichen section** Courtesy of Ralf Wagner; **puffer** Andrey Nekrasov/Pixtal/AGE Fotostock; **Azolla** Daniel L Nickrent; **37.9** ant Juan Carlos Vindas/Moment Open/Getty Images; **fungal garden** Martin Dohrn/Nature Picture Library; **root** Yoshihiro Kobae; **nectar** Oxford Scientific/Getty Images; **37.10** Sarah Lydia Lebeis; **37.12** Scimat/Science Source; **37.14** sheath Hugues B. Massicotte/University of Northern British Columbia Ecosystem and Management Program, Prince George, BC, Canada; **cells, arbuscules** Mark Brendrett; **37.15** fern David Wall/Alamy Stock Photo; **mistletoe** Peter Lane/Alamy Stock Photo; **dodder** Emilio Erez/Alamy Stock Photo; **Indian pipe** Martin Shields/Alamy Stock Photo; **pitcher plants** Dorling Kindersley Ltd/Alamy Stock Photo; **ant on pitcher plant** Paul Zahl/Science Source; **sundew** Fritz Polking/Frank Lane Picture Agency Limited W. Rolles/Arco Images GmbH/Alamy Stock Photo; **Venus flytrap** Chris Mattison/Nature Picture Library; **p. 821** footprint Mode Images/Alamy Stock Photo.

Chapter 38 **38.1** blickwinkel/Alamy Stock Photo; **inset** Nicolas J. Vereecken; **38.4** hazel carpellate Friedhelm Adam/imageBROKER/Getty Images; **hazel staminate** Wildlife GmbH/Alamy Stock Photo; **dandelions** © Björn Rørslett/NN/Samfoto/Sipa USA; **moth** Doug Backlund/WildPhotosPhotography.com; **blowfly** Kjell B. Sandved/Science Source; **bat** Rolf Nussbaumer/imageBROKER/AGE Fotostock; **hummingbird** Rolf Nussbaumer/Nature Picture Library; **38.5** W. Barthlott and W. Rauh/Nees Institute for Biodiversity of Plants; **38.6 top** Michael Clayton/Botany Dept., University of Wisconsin; **bottom** Ed Reschke/Photolibary/Getty Images; **38.10** Blickwinkel/Alamy Stock Photo; **38.12** coconut Kevin Schafer/Alamy Stock Photo; **Alsomitra macrocarpa** Aquiya/Fotolia; **dandelion** Steve Bloom Images/Alamy Stock Photo; **maple seed** Chrispo/Fotolia; **tumbleweed** Nurlan Kalchinov/Alamy Stock Photo; **Tribulus terrestris** California Department of Food and Agriculture's Plant Health and Pest Prevention Services; **squirrel** Alan Williams/Alamy Stock Photo; **feces** Kim A. Cabrera; **ant** Benoit Guénard; **38.13** Dennis Frates/Alamy Stock Photo; **Scientific Skills Exercise** monkeyflower Ken Barber/Alamy Stock Photo; **hummingbird** Dec Hogan/Shutterstock; **38.14a** Marcel Dorken; **38.14b** Nobumitsu Kawakubo; **38.15** Meriel G. Jones, University of Liverpool School of Biological Sciences; **38.16** Dorling Kindersley Ltd/Alamy Stock Photo; **p. 837** Herrera-Estrella Michael Starghill, Michael Starghill Photography; **38.17** Gary P. Munkvold; **p. 838** Gonsalves Courtesy of Dennis Gonsalves; **38.18** ton koene/Alamy Stock Photo; **p. 841** pollen Dartmouth College Electron Microscope Facility.

Chapter 39 **39.1** Christopher Ison/Alamy Stock Photo; **39.2** Natalie Bronstein; **39.6** From: Regulation of polar auxin transport by ATPIN1 in *Arabidopsis* vascular tissue. L. Gälweiler et al. *Science.* 1998 Dec 18;282(5397):2226-30; Fig. 4; **39.9a** Richard Amasino; **39.9b** Fred Jensen, Kearney Agricultural Center; **39.11** left Mia Molvray; **right** Karen E. Koch; **39.13a** Kurt Stepnitz; **39.13b** Joseph J. Kieber; **39.14** Ed Reschke; **39.16** Nigel Cattlin/Alamy Stock Photo; **39.18** Martin Shields/Alamy Stock Photo; **p. 858** Satter Robin Heyden/Pearson Education; **39.22** Michael L. Evans/Ohio State University; **39.23** From the cover of *Cell*, Volume 60, Issue 3, 9 February 1990, Janet Braam, Ronald W. Davis. Used by permission, Copyright ©1990 Cell Press. Image